

Big Data & NoSQL Databases, Spring 2026
Project
Due date is Friday, May 22nd, 2026 at 11:59 PM
Submitted in groups of 4 up to 6 (No exceptions)
(Can be cross tutorial)

You are a lead big data engineer for a city planning agency in New York. You have been provided with historical **Citi Bike trip data**. The dataset provides comprehensive ride data, including details such as start and end times, start and end station locations, and bike identifiers. Your task is to process this high-volume data to understand who is using the system, predict trip patterns, and provide recommendations for station expansions based on user demographics and trip behavior.

Since the dataset is relatively large, it is recommended to load and read the data directly from a shared drive link rather than uploading it locally.

Requirements

1. Data Cleaning & Engineering:

- a) Initialize a Spark session and load the CSV file into a Spark DataFrame.
- b) Begin by exploring the dataset and identify any anomalies or inconsistencies. You are expected to clean and prepare the data. Some values may be missing explicitly (e.g., NULL) or implicitly (e.g., zeroes or empty strings). Handle such cases thoughtfully in a way that allows for reliable querying and analysis.
- c) You are expected to create new columns that provide useful insights:
 - **Rider Age**: Derive an Age column using **Year of Birth**
 - **Trip Duration**: Calculate the duration in seconds.
 - **Trip Distance**: Calculate using the Haversine formula based on start and end station coordinates to estimate real travel distance.
 - **Trip Speed** (Km/h): Calculate an estimated speed using the trip distance and trip duration (For any conversions, utilize the spark UDFs).
 - **Time Categorization**: Create a Period_of_Day feature (Morning, Afternoon, Evening, Night).
 - **Start Month**: Extracted from the start timestamp.
And make sure to validate your dataset after each modification.
- d) You are expected to identify and flag records that meet any of the following noise criteria:
 - **Trips with a trip duration of less than 60 seconds.**
 - Trips where the calculated speed (using Station Lat/Long and Trip Duration) exceeds 40 km/h, suggesting a system glitch rather than a human-powered ride.
 - Records where the year of birth results in a rider age of over 100 years or under 12 years.
 - Any rows missing essential identifiers such as Start Station Name, End Station Name, or Bike ID.

2. Analytical Queries:

Write the following queries that provide different insights into the data. You may implement them using **Spark SQL** or **Spark DataFrame API**.

For each query, you must provide a brief business interpretation explaining what the result means and how it could support decision-making.

- a) What percentage of trips are round trips (start station = end station) per user type?
- b) Which are the most popular start stations overall, and how does their usage rank?
- c) Which hours of the day experience the highest demand (rush hours)?
- d) Create an age-group feature using a UDF that classifies users into predefined categories such as Young, Adult, or Senior based on age intervals and insert it as a new column. Then, analyze how trip duration varies across the different age groups.
- e) Analyze how bike usage and trip behavior change across different seasons. Create a Season feature using a UDF that classifies each trip into Winter, Spring, Summer, or Autumn based on the ride month, register it for use in Spark queries, and insert it as a new column in the dataset.
- f) Determine if specific Bike IDs are being over-utilized by counting their total trip seconds and identifying those needing maintenance.
- g) Compare the most popular end station names for subscribers and customers. Analyze differences in geographic mobility patterns, including whether subscribers concentrate around residential or business areas while customers are more associated with recreational or tourist locations.
- h) Identify the station pairs (Start Station - End Station) with the highest trip demand based on the average number of rides between them.
- i) Identify whether there are statistically significant differences in riding behavior across genders, focusing on metrics such as trip speed and trip duration.
- j) Analyze how trip behavior differs between weekday and weekend rides by comparing average trip duration, average distance traveled, and average riding speed.

3. SparkML Task:

- Use **SparkML** to build three machine learning models to predict the gender of the rider. This includes normalizing the necessary columns, selecting the most relevant features, and assembling them into a single feature vector using **VectorAssembler**.
- Split the dataset into training and testing sets, and evaluate the accuracy of each model on the test data.
- Train and compare at least three different classifiers (e.g., logistic regression, decision tree, and random forest).
- Summarize the results and discuss which model performs best and why. Additionally, highlight the most influential features in predicting gender, if supported by the chosen models.

Bonus (Dashboard Visualization)

Create an interactive dashboard using the results of your analytical queries (e.g., Power BI, Tableau, Bokeh or Spark-integrated visualization libraries).

The dashboard should summarize key insights from the Citi Bike dataset and include:

- **Temporal analysis:** Hourly demand, weekday vs weekend trends, and seasonal variation.
- **Station analysis:** Most popular start or end stations and top station pairs.
- **User behavior:** Trip duration and riding patterns across age groups, genders, and user types.
- **Bike utilization:** Identification of over-used bikes and distribution of trip durations per bike.
- **Requirements:**
 - At least 4 visualizations
 - At least 1 interactive filter (e.g., time, user type, station)
 - All visuals must be based on Spark query outputs

<i>Column</i>	<i>Description</i>
bike_id	A unique identifier for each ride. This column helps distinguish between different rides in the dataset.
starttime	The date and time when the ride started. This is useful for analyzing ride patterns across different times of the day, week, or year.
stoptime	The date and time when the ride ended. Combined with the start time, it helps calculate ride duration.
start_station_name	The name of the station where the ride started. This helps analyze station popularity and usage patterns.
start_station_id	A unique identifier for the start station, ensuring precise station identification.
end_station_name	The name of the station where the ride ended. This helps analyze destination patterns and station usage.
end_station_id	A unique identifier for the end station, ensuring accurate station identification.
start_station_lat	The latitude coordinate of the start station, useful for mapping ride starting locations geographically.
start_station_longitude	The longitude coordinate of the start station, used together with latitude for geographic analysis.
end_station_latitude	The latitude coordinate of the end station, useful for mapping ride ending locations geographically.
end_station_lng	The longitude coordinate of the end station, used together with latitude for geographic analysis.
gender	Indicates the gender of the user (0 = Unknown, 1 = Male, 2 = Female).
user_type	Indicates whether the rider is a subscribed member or a customer user, helping analyze differences in user behavior and ride patterns.

Tabelle 1: Citi Bike Dataset Schema

Deliverables

- Your source code notebook should fully implement all the specifications listed above. **Ensure that all outputs, results, and visualizations are included in the notebook before submission**, as the notebook may be evaluated directly without re-running the code. Missing outputs may result in grade deductions.
- Part of the grade will be on how readable your code is. Use explanatory comments whenever possible
- Your code is to be submitted to **bigdata602.25@gmail.com** as a zip file containing all your work with **subject:** “Project” and IDs of the team members in the body of the email.
- You will be evaluated based on the submitted notebook before the deadline only.

Note: You will be asked for the reasoning of any actions, queries or decisions you have taken in the implementation of this project during the evaluations that have led to your answers/results
